

Lecture 1

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1 Invertible Matrix Theorem

1.1 Definition

Let A be a square $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A , the statements are either all true or all false.

1. A is invertible.
2. A is row equivalent to I .
3. A has n pivot positions.
4. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
5. The columns of A are linearly independent.

6. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
7. The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
8. The columns of A span $\mathbb{R}^n$.
9. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
10. There is an $n \times n$ matrix C such that $CA = I$.
11. There is an $n \times n$ matrix D such that $AD = I$.
12. A^T is invertible.
13. The columns of A form a basis of $\mathbb{R}^n$.
14. $\text{Col } A = \mathbb{R}^n$.
15. $\dim \text{Col } A = n$.
16. $\text{rank } A = n$.
17. $\text{Nul } A = \{0\}$.
18. $\dim \text{Nul } A = 0$.
19. The number 0 is *not* an eigenvalue of A .
20. $\det A \neq 0$.
21. $(\text{Col } A)^\perp = \{0\}$.
22. $(\text{Nul } A)^\perp = \mathbb{R}^n$.
23. $\text{Row } A = \mathbb{R}^n$.

1.2 Exercise

Thm 9.1.16 Let $A \in M_n$ and let $\lambda \in \mathbb{C}$. Then the following statements are equivalent:

- (a) λ is an eigenvalue of A .
- (b) $A\mathbf{x} = \lambda\mathbf{x}$ for some nonzero $\mathbf{x} \in \mathbb{C}^n$.

- (c) $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, that is, $\text{nullity}(A - \lambda I) > 0$.
- (d) $\text{rank}(A - \lambda I) < n$.
- (e) $A - \lambda I$ is not invertible.
- (f) $A^\top - \lambda I$ is not invertible.
- (g) λ is an eigenvalue of $A^\top$.

Proof

(a) $\Leftrightarrow$ (b)

By definition 9.1.1, if $A\mathbf{x} = \lambda\mathbf{x}$ and $\mathbf{x} \neq \mathbf{0}$ then $(\lambda, \mathbf{x})$ is an eigenpair of A , meaning λ is an eigenvalue of A .

(b) $\Leftrightarrow$ (c)

These are restatements of one another.

(c) $\Leftrightarrow$ (d)

By rank nullity theorem, if $\text{nullity}(A - \lambda I) > 0$, then $\text{rank}(A - \lambda I) < n$.

(d) $\Leftrightarrow$ (e)

By property 1 and 16 of IMT, if $\text{rank}(A - \lambda I) < n$ then $(A - \lambda I)$ is not invertible.

(e) $\Leftrightarrow$ (f)

Suppose we have $A - \lambda I = \begin{bmatrix} a - \lambda I & b \\ c & d - \lambda I \end{bmatrix}$, and suppose it is not invertible, its determinant is $(a - \lambda I)(d - \lambda I) - bc = 0$. Consider the case of $A^\top - \lambda I$, we have $\begin{bmatrix} a - \lambda I & c \\ b & d - \lambda I \end{bmatrix}$, whose determinant is $(a - \lambda I)(d - \lambda I) - cb$. And by commutativity of multiplication, $bc = cb$, so the determinant is also 0, meaning $A^\top - \lambda I$ is also not invertible.

(f) $\Leftrightarrow$ (g)

Since (a) is equivalent to (e), the same is true for $A^\top$.

2 Annihilator

2.1 Some Roots of an Annihilator are Eigenvalues

While all the eigenvalues of a matrix are roots of each of its annihilators, the roots of an annihilator are not necessarily eigenvalues.

This raises a question:

Must some of the roots of an annihilator of a matrix be eigenvalues of the matrix? The answer is yes.

Let $p(z) = a_0 + a_1z + a_2z^2 + \dots + a_kz^k$ be an annihilator of a matrix $A \in \mathbb{M}_n$. Factor p as follows: $p(z) = a_k(z - \lambda_1)(z - \lambda_2) \dots (z - \lambda_k)$, $a_k \neq 0$. Since p annihilates A , we have

$$0 = p(A) = a_k(A - \lambda_1 I)(A - \lambda_2 I) \dots (A - \lambda_k I)$$

Since the zero matrix is not invertible, one of the factors $A - \lambda_i I$ is not invertible. This ensures that λ_i is an eigenvalue of A .

2.2 Each Matrix has an Eigenvector

2.2.1 Proof 1

λ is an eigenvalue of $A \in \mathbb{M}_n$ iff $\det(A - \lambda I) = 0$.

$p(\lambda) = \det(A - \lambda I)$ is a polynomial of degree n (the characteristic polynomial of A).

Therefore, λ is an eigenvalue of A iff λ is a root of p (i.e. p)

By the Fundamental Theorem of Algebra (FTA), each polynomial over the complex has a root.

Hence, each matrix $A \in \mathbb{M}_n$ has an eigenvalue.

2.2.2 Proof 2

We've shown that each matrix $A \in \mathbb{M}_n$ has an annihilator p .

By the FTA, p must have a root.

In 2.1, we showed that at least one root of p is an eigenvalue of A .

Hence, each matrix $A \in \mathbb{M}_n$ has an eigenvalue.